module 2 random forest

```
from google.colab import files
uploaded = files.upload()
```

Choose Files student-mat (2).csv
student-mat (2).csv(text/csv) - 56993 bytes, last modified: 3/18/2026 - 100% done

```
Saving student-mat (2).csv to student-mat (2).csv
```

```
import pandas as pd

df = pd.read_csv("student-mat (2).csv")
```

```
print(df.columns)
```

```
Index(['school;sex;age;address;famsize;Pstatus;Medu;Fedu;Mjob;Fjob;reason;guardian;traveltime;studytime;failures;schoolsup;famsu
```

```
df = pd.read_csv("student-mat (2).csv", sep=';')
```

```
print(df.columns)
```

```
Index(['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu', 'Fedu',
       'Mjob', 'Fjob', 'reason', 'guardian', 'traveltime', 'studytime',
       'failures', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery',
       'higher', 'internet', 'romantic', 'famrel', 'freetime', 'goout', 'Dalc',
       'Walc', 'health', 'absences', 'G1', 'G2', 'G3'],
      dtype='object')
```

```
X = df.drop("G3", axis=1)
y = df["G3"]
```

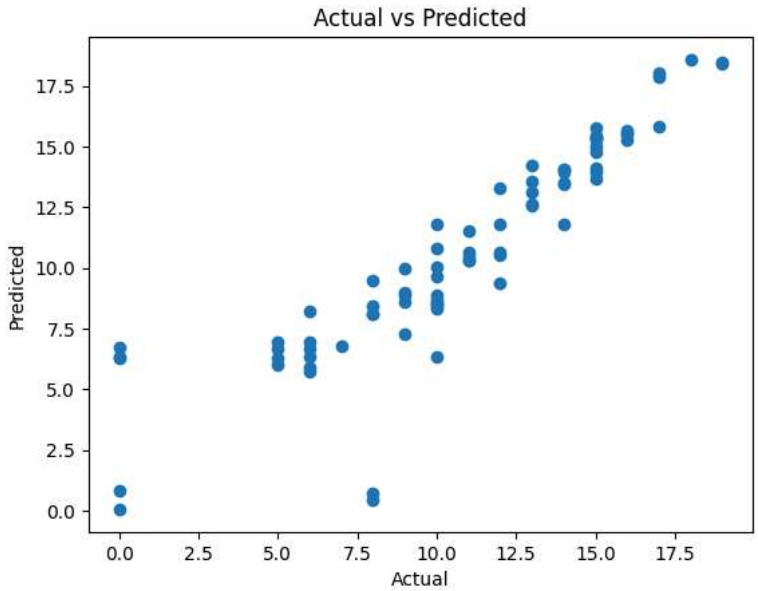
```
df = df.dropna(axis=1, how='all')
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset (IMPORTANT: sep=';')
df = pd.read_csv("student-mat (2).csv", sep=';')

# Convert categorical → numeric
df = pd.get_dummies(df, drop_first=True)

# Features & Target
X = df.drop("G3", axis=1)
y = df["G3"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model
model = RandomForestRegressor()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
MSE: 4.030451898734177
R2 Score: 0.8034409847466634
```

```
import matplotlib.pyplot as plt

plt.scatter(y_test, y_pred)
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Actual vs Predicted")
plt.show()
```



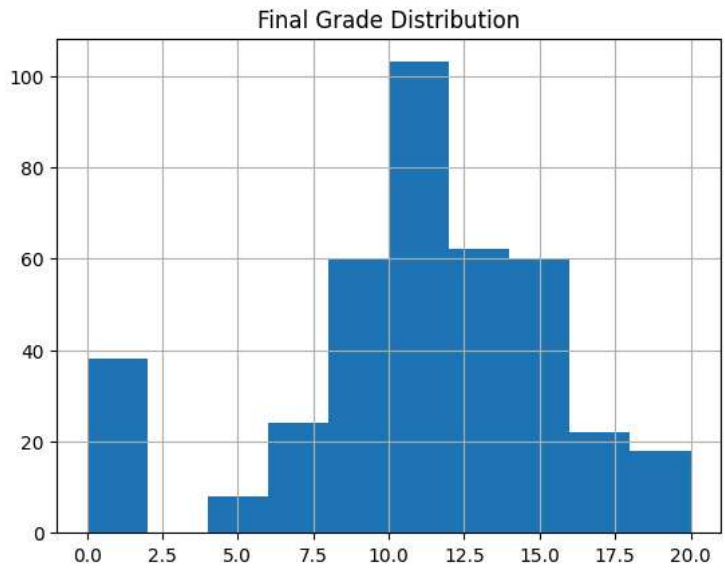

```
print(df.head())
print(df.info())
print(df.describe())
```

```
min      1.000000    1.000000    1.000000    1.000000    1.000000    1.000000
25%      4.000000    3.000000    2.000000    1.000000    1.000000    3.000000
50%      4.000000    3.000000    3.000000    1.000000    2.000000    4.000000
75%      5.000000    4.000000    4.000000    2.000000    3.000000    5.000000
max      5.000000    5.000000    5.000000    5.000000    5.000000    5.000000

         absences          G1          G2          G3
count  395.000000  395.000000  395.000000  395.000000
mean     5.708861   10.908861   10.713924   10.415190
std      8.003096    3.319195    3.761505    4.581443
min      0.000000    3.000000    0.000000    0.000000
25%      0.000000    8.000000    9.000000    8.000000
50%      4.000000   11.000000   11.000000   11.000000
75%      8.000000   13.000000   13.000000   14.000000
max     75.000000   19.000000   19.000000   20.000000
```

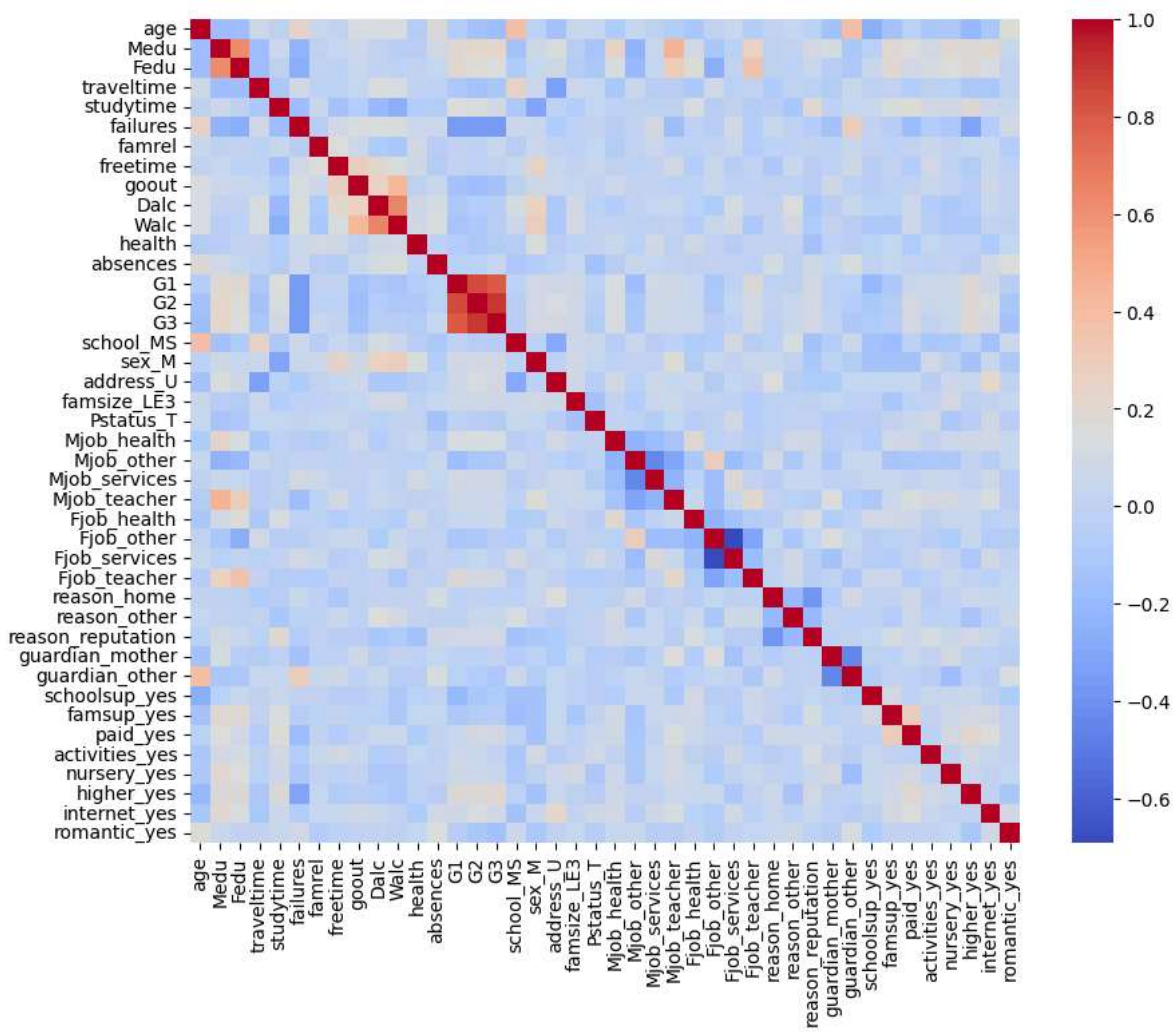
```
print(df.isnull().sum())
```

```
age                  0
Medu                 0
Fedu                 0
traveltime           0
studytime            0
failures             0
famrel               0
freetime             0
goout                0
Dalc                 0
Walc                 0
health               0
absences             0
G1                   0
G2                   0
G3                   0
school_MS            0
sex_M                0
address_U            0
famsize_LE3          0
Pstatus_T            0
Mjob_health          0
Mjob_other           0
Mjob_services        0
Mjob_teacher         0
Fjob_health          0
Fjob_other           0
Fjob_services        0
Fjob_teacher         0
reason_home          0
reason_other         0
reason_reputation    0
guardian_mother      0
guardian_other       0
schoolsup_yes        0
famsup_yes           0
paid_yes             0
activities_yes       0
nursery_yes          0
higher_yes           0
internet_yes         0
romantic_yes         0
dtype: int64
```

```
import matplotlib.pyplot as plt

# Histogram
df['G3'].hist()
plt.title("Final Grade Distribution")
plt.show()
```



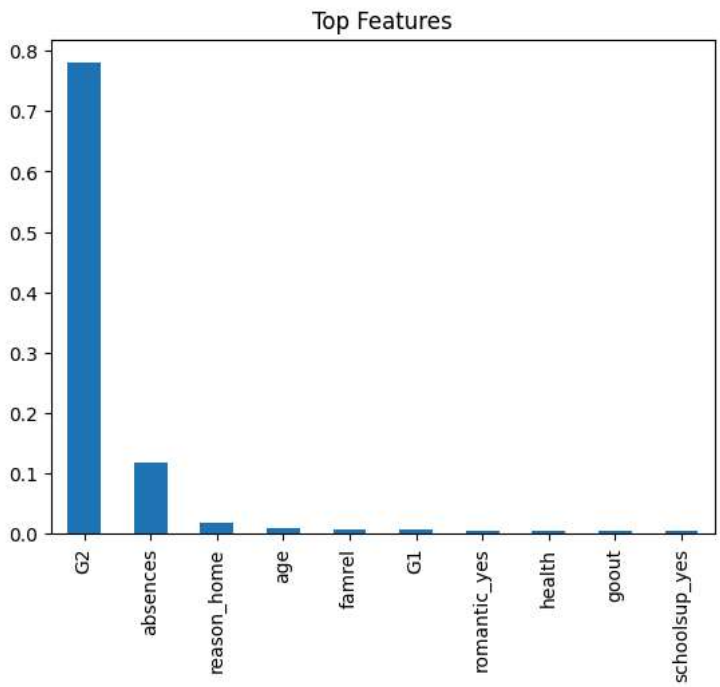

```
import seaborn as sns

plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), cmap='coolwarm')
plt.show()
```




```
model = RandomForestRegressor(n_estimators=200, max_depth=10)
model.fit(X_train, y_train)
```

```
RandomForestRegressor
RandomForestRegressor(max_depth=10, n_estimators=200)
```

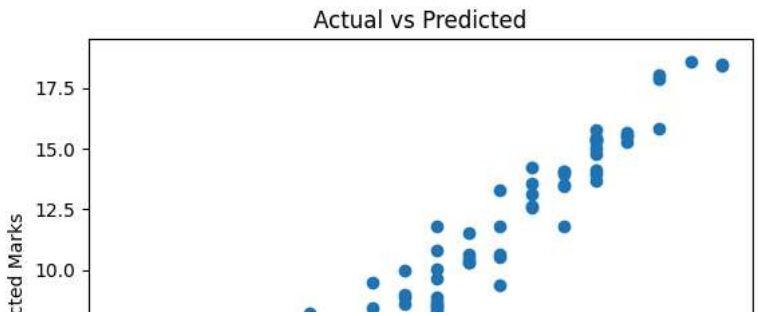
```
import pandas as pd

importance = pd.Series(model.feature_importances_, index=X.columns)
importance.sort_values(ascending=False).head(10).plot(kind='bar')
plt.title("Top Features")
plt.show()
```




```
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
MSE: 4.030451898734177
R2 Score: 0.8034409847466634
```

```
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Marks")
plt.ylabel("Predicted Marks")
plt.title("Actual vs Predicted")
plt.show()
```


Actual vs Predicted
17.5
15.0
12.5
10.0
cted Marks